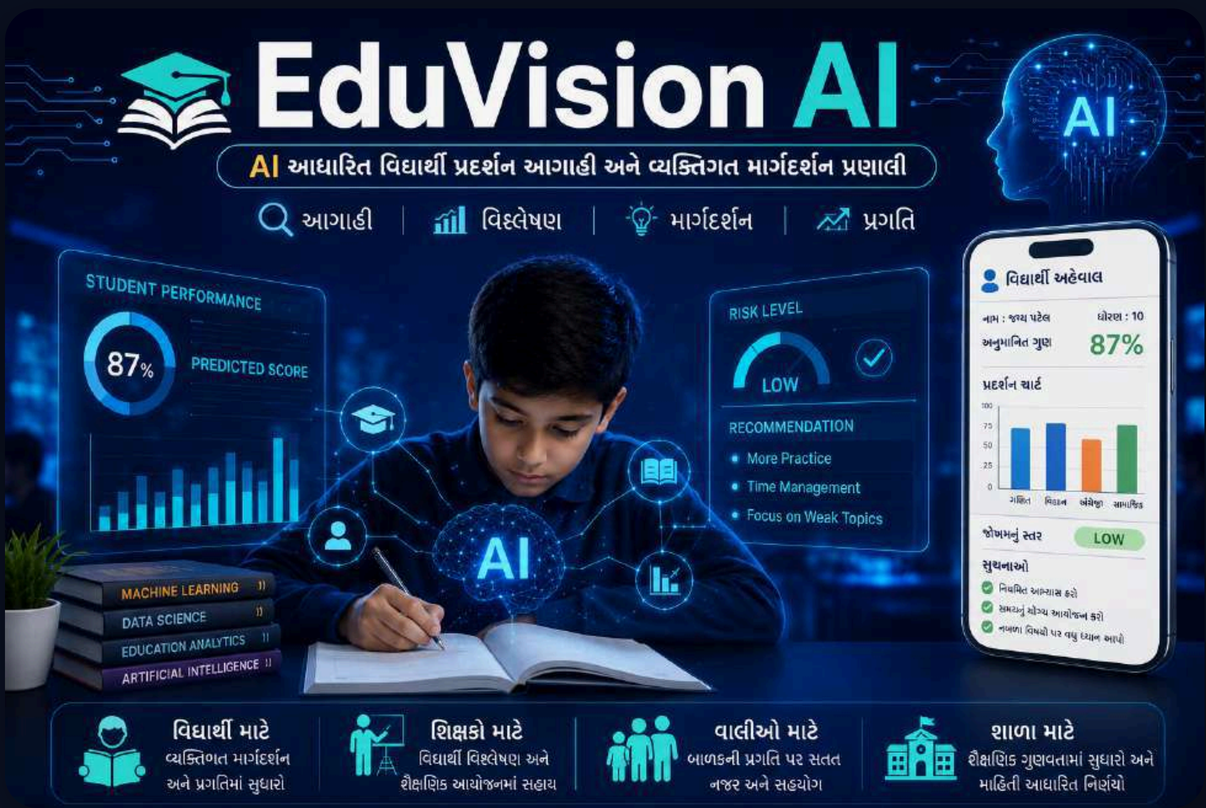
🚀 SCIENCE FAIR 2026 EDITION

What if AI could predict academic failure before it happens?

Traditional report cards tell parents what went wrong after final exams. Here is how we engineered an intelligent, early-warning machine learning platform to fix education proactively.



By Yash Patel · Mentored by Shri Manojbhai Parmar

THE CORE PROBLEM

Why standard educational grading is fatally broken:

Teachers manage 40+ students per classroom. Recognizing silent indicators of failure before board examinations is humanly impossible without data.

❌ **01**

Post-Mortem Interventions

Corrective tutoring only happens after a student flunks, leaving zero recovery time before progression.

❌ **02**

Invisible Behavioral Factors

Absences and study routines quietly degrade comprehension weeks before any test paper is written.

❌ **03**

Generic Generic Advice

"Work harder" is useless advice without specific mathematical targets on study hours and syllabus focus.

💡 **The Fix**

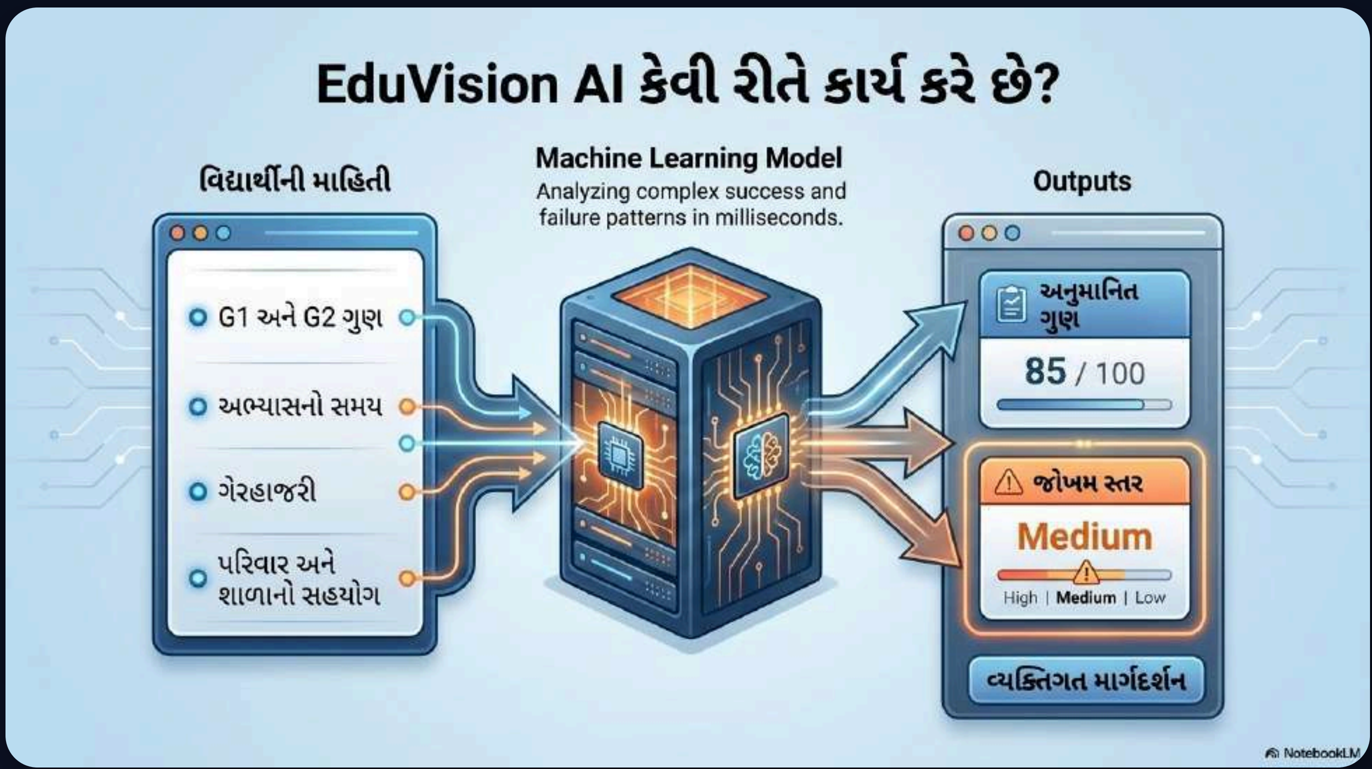
Predictive AI Mentorship

Early identification through machine learning regression with explainable feature weights.

INTELLIGENT EDUCATIONAL AI

Meet EduVision AI: Proactive Mentorship

Trained on 395 real student longitudinal records (UCI Benchmark), predicting final grades (G3) out of 20 with over 81% precision.



MACHINE LEARNING RIGOR

Why Gradient Boosting outperformed all rivals:

We trained and cross-validated multiple regression models. Gradient Boosting iteratively learns from previous tree residuals, minimizing variance.

Linear Regression	MAE: 1.4429	R^2: 0.7690
Random Forest Regressor	MAE: 1.2061	R^2: 0.8132
🏆 **Gradient Boosting (EduVision)**	**MAE: 1.1804**	**R^2: 0.8138**

UI/UX INNOVATION

Zero-Collision HUD Gauge Meter

Designed for Science Fair projection with a space-oriented dark/light theme, layered SVG architecture, and instant explainable feedback.

01

Dynamic Progress Fill

Active neon arc illuminates precisely to student percentage with zero line-crossing bugs.

02

100% Gujarati Localization

Rendered with modern regional geometric fonts (Anek Gujarati & Noto Sans Gujarati).

INTERACTIVE EXPLORATION

The "What-If" Academic Simulator

Students don't just see a score—they discover how to change it. Moving the sliders recalibrates predictions in real-time.

"What if I study 2 hours more?"

Instantly recalculates regression delta (+1.8 marks increase) showing immediate positive reinforcement.

"What if attendance hits 95%?"

Maps the non-linear relationship between absenteeism and examination confidence.

PRINT & PDF ENGINE

Instant 1-Page Official Report Card

At the Science Fair kiosk, students and parents generate a verified, single-page A4 certificate to take home.

📄

Isolated Print Portal

React Portal & strict @media print rules eliminate background web UI, outputting pure report cards.

✍️

Verified School Crest & Signatures

Authorized slots for Project Mentor (Shri Manojbhai Parmar) and Parent endorsement.

100% LIVE & OPEN SOURCE

Experience EduVision AI 2.0 Today!

Deployed on Vercel Edge with zero latency. Try the live demo, review the code, and let me know your thoughts!

🌐 **Live App:** https://eduvision-ai-kappa.vercel.app

💻 **GitHub:** https://github.com/yashintelligence/eduvision-ai

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